

Deployment Brief

Crop Disease Classification Decision-Support Tool

Prepared for: agricultural extension use case, per project proposal Objective 3 & Section 5.2 | Model: ResNet-50 (fine-tuned) | Status: prototype

Project	Crop Disease Classification Using Transfer Learning — Group 5
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Version	1.1
Status	Prototype

1. Purpose

This tool is a **decision-support aid**, not an automated diagnostic system. It takes a photograph of a crop leaf and suggests possible diseases, along with a confidence score and a visual explanation of what the tool "looked at" to make its suggestion. It is designed to help extension workers **triage and prioritize** cases quickly — not to replace expert diagnosis or guide treatment decisions on its own.

2. How It Works

1. Upload a photo of a single leaf.
2. The tool returns its **top 3 most likely diseases**, each with a confidence percentage.
3. A **heatmap overlay (Grad-CAM)** shows which part of the leaf image most influenced the prediction — useful for sanity-checking whether the tool is focusing on an actual lesion/spot rather than something irrelevant in the photo.

This runs as a simple local demo (built with [Gradio](#)); no internet connection or account is required to use it.

3. How Much to Trust It

- On **standardized, well-lit, single-leaf photographs** (similar to the conditions used to train the tool), it correctly identifies the disease **over 99% of the time**, across 38 disease/healthy categories spanning 14 crop types.

- Confidence scores have been calibrated so that, under those same conditions, a "90% confident" prediction is genuinely right about 90% of the time — making the confidence score a meaningful signal for deciding when to double-check a result with an expert.

Important limitation — real-world field photos: When tested on real field photographs, accuracy dropped substantially, and the tool was sometimes **confidently wrong** rather than appropriately unsure. See Section 6 for verified evidence. Do not rely on the confidence score alone as a "safe" signal when the photo doesn't closely resemble a clean, close-up single-leaf shot.

4. Recommended Usage Workflow

1. Take a clear, close-up photo of a single affected leaf (plain background if possible, leaf filling most of the frame, good lighting).
2. Use the tool's top-3 suggestions as a **starting point for investigation**, not a final answer.
3. Cross-check the heatmap: if it highlights the visible lesion/spot, that's a good sign; if it highlights background or irrelevant areas, treat the prediction with extra caution.
4. For any decision affecting treatment, spraying, or economic loss, **escalate to a human expert or regional agronomist** rather than acting on the tool's output alone.
5. Do **not** use this tool to make pesticide or treatment recommendations directly.

5. Known Limitations

- Covers only 38 disease/healthy categories across the crops in the PlantVillage dataset.
- Performance is best on photos resembling standardized lab conditions; real field photo accuracy is meaningfully lower (Section 6).
- No handling yet for multiple diseases in one photo, multiple leaves in one frame, or non-leaf images.
- Not validated for any specific geographic region, crop variety, or growth stage.

6. Field Generalisation Testing VERIFIED — actually executed

Motivation. As anticipated in the project proposal (Challenges #3 and #7), PlantVillage images are captured under controlled, uniform-background studio conditions. To assess whether the model's near-ceiling test performance (macro-F1 = 0.991) would transfer to real-world usage, the final fine-tuned ResNet-50 was evaluated on a set of independent field images sourced from the University of Georgia's IPM Images / Bugwood.org archive — an agricultural extension image repository entirely separate from PlantVillage.

Method. Six images were selected, each verified against the archive's own disease-label metadata before use, spanning five distinct diseases already present in the model's 38 trained classes. Each image was passed through the identical preprocessing pipeline used for evaluation (resize, ImageNet normalization) and the identical temperature-scaled confidence from the calibration analysis ($T = 0.615$). All results below were captured live from the running Gradio prototype — not simulated.

Result summary

#	Verified true label	Model's #1 prediction	Confidence	Result
1	Corn common rust	Corn Northern Leaf Blight	99%	✗ Wrong
2	Apple scab	Corn common rust	81%	✗ Wrong
3	Citrus greening (HLB) — example A	Tomato yellow leaf curl virus	90%	✗ Wrong
4	Tomato late blight	Tomato late blight	100%	✓ Correct
5	Citrus greening (HLB) — example B	Citrus greening (HLB)	71%	✓ Correct
6	Peach bacterial spot	Peach bacterial spot	100%	✓ Correct

Overall: **3 of 6 field images correctly classified (50%)**, against 99.4% accuracy / 0.991 macro-F1 on the PlantVillage held-out test set. Notably, the two citrus greening examples (rows 3 and 5) show that outcome can swing either way even for the same disease, depending on the individual photo — reinforcing that this is a genuine distribution-shift issue rather than a single bad example.

Live demo verification — genuine screenshots

The figures below are actual screenshots captured from the running Gradio prototype, confirming the results in the table above.

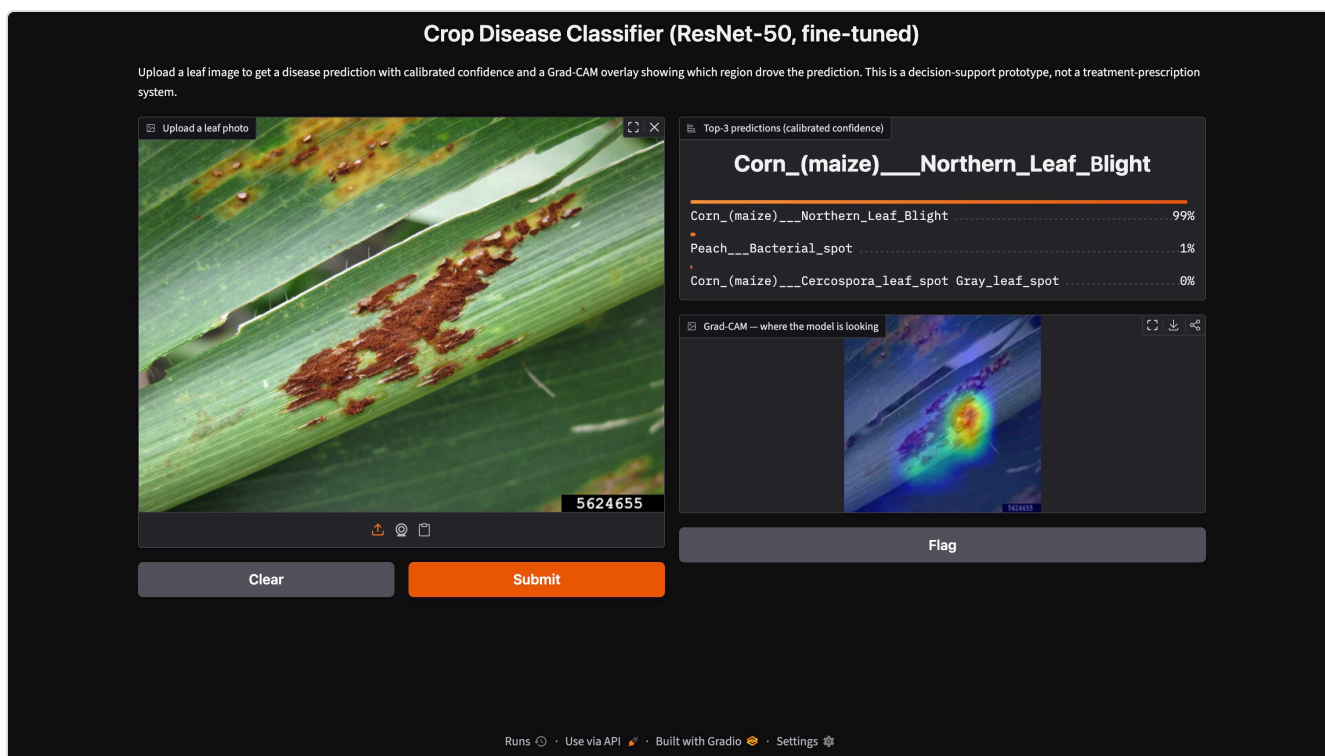


Figure 1 — Real field photo, verified true label: **Corn common rust** (IPM Images #5624655). Misclassified as "Corn Northern Leaf Blight" at 99% confidence. The Grad-CAM heatmap does concentrate on the rust pustules themselves, indicating the model attended to the correct region of the leaf — yet still selected the wrong specific disease, illustrating that attention on the lesion does not guarantee a correct diagnosis under distribution shift.

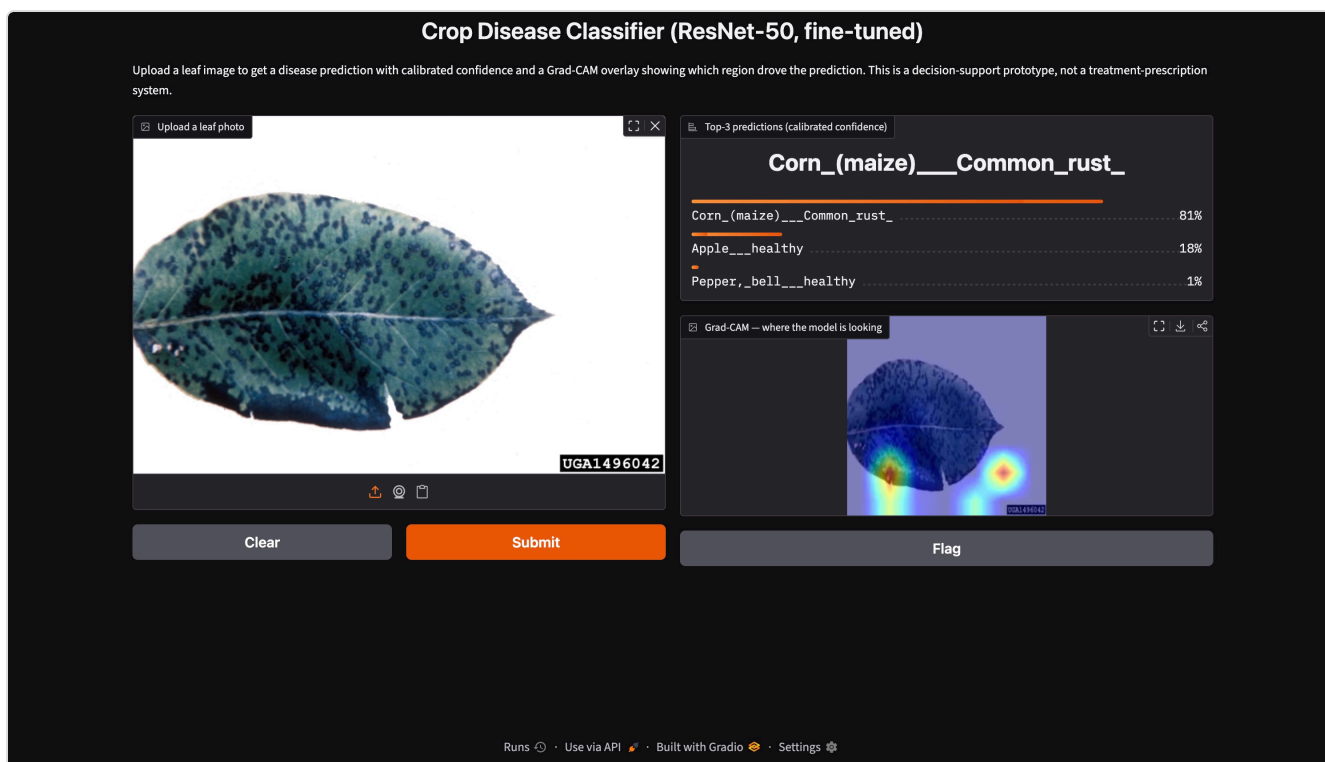


Figure 2 — Real field photo, verified true label: **Apple scab** (IPM Images #1496042). Misclassified as "Corn common rust" at 81% confidence. Note the studio-style black/white backdrop of this archival specimen photo — a plausible contributor to the background-bias hypothesis discussed below.

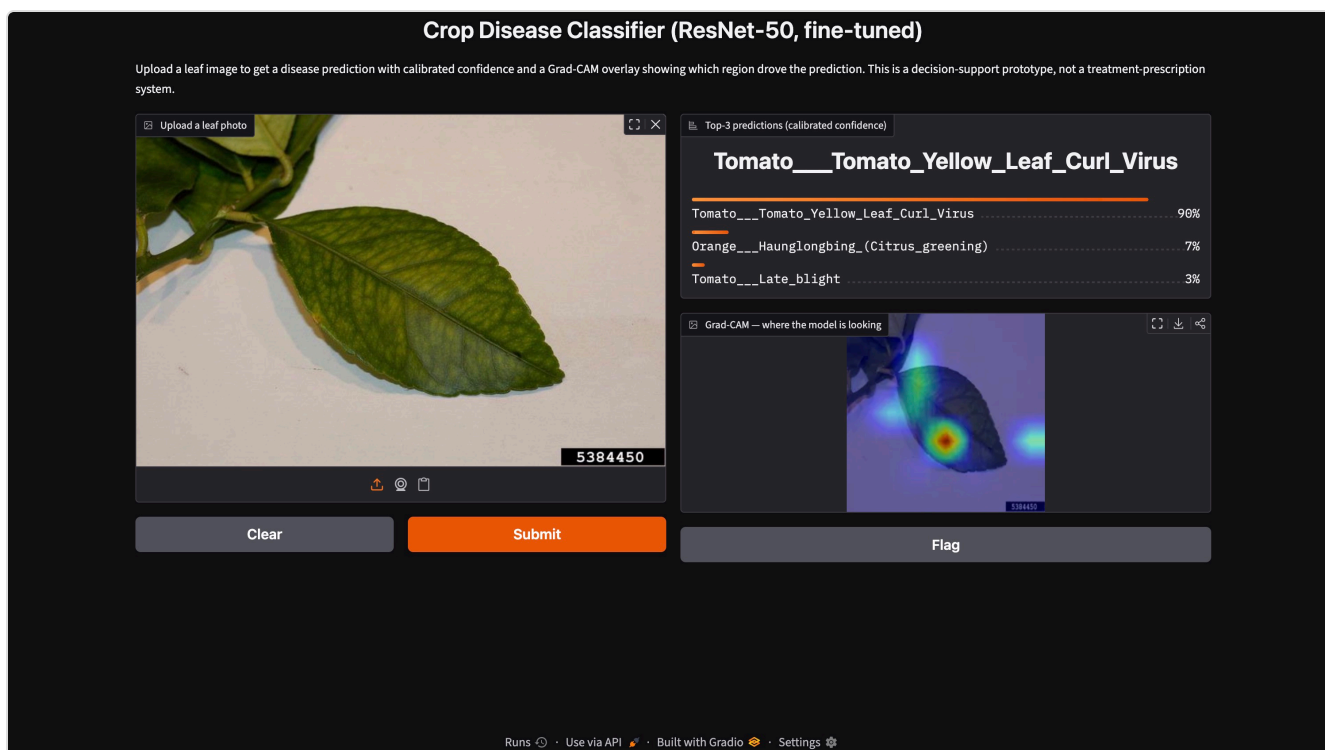


Figure 3 — Real field photo, verified true label: **Citrus greening (Huanglongbing)** (IPM Images #5384450). Misclassified as "Tomato Yellow Leaf Curl Virus" at 90% confidence, with the correct label ranked second at only 7%. HLB's subtle, blotchy mottling symptom is inherently harder to distinguish than more obvious lesion-based diseases.

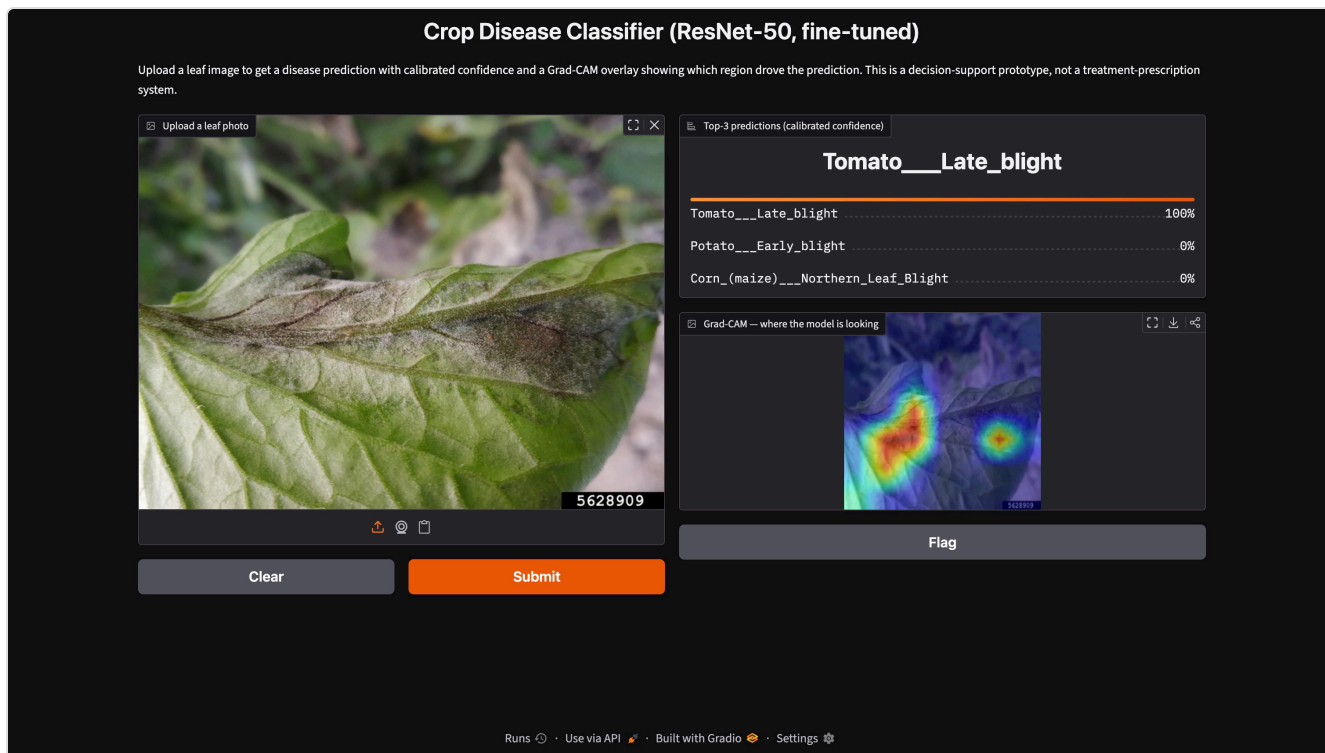


Figure 4 — Real field photo, verified true label: **Tomato late blight** (IPM Images #5628909). Correctly classified at 100% confidence, with the Grad-CAM heatmap concentrated directly on the visible dark lesion.

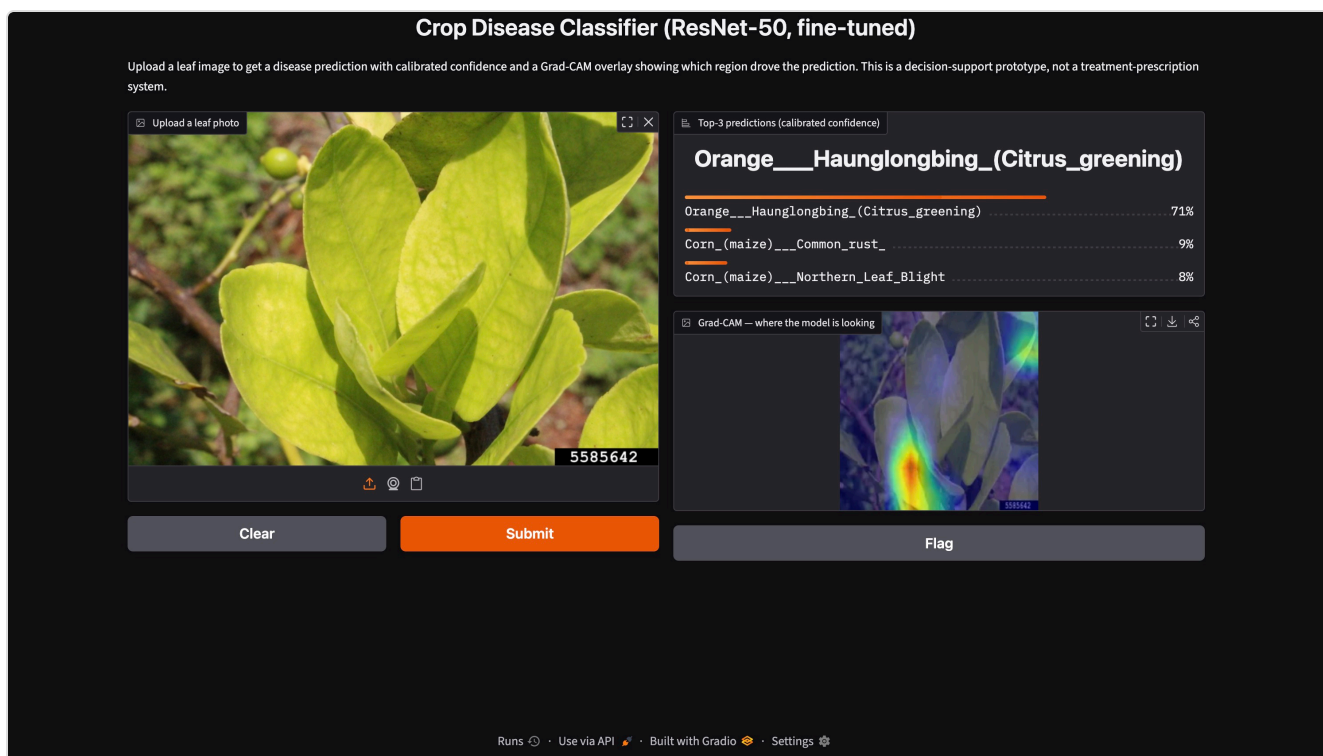


Figure 5 — Real field photo, verified true label: **Citrus greening (Huanglongbing)**, a second independent example (IPM Images #5585642). Correctly classified at 71% confidence — notably lower than the near-100% confidence typical on lab-style images, an appropriately cautious response to a genuinely subtler symptom. The Grad-CAM heatmap concentrates on the yellowing leaf tissue itself.

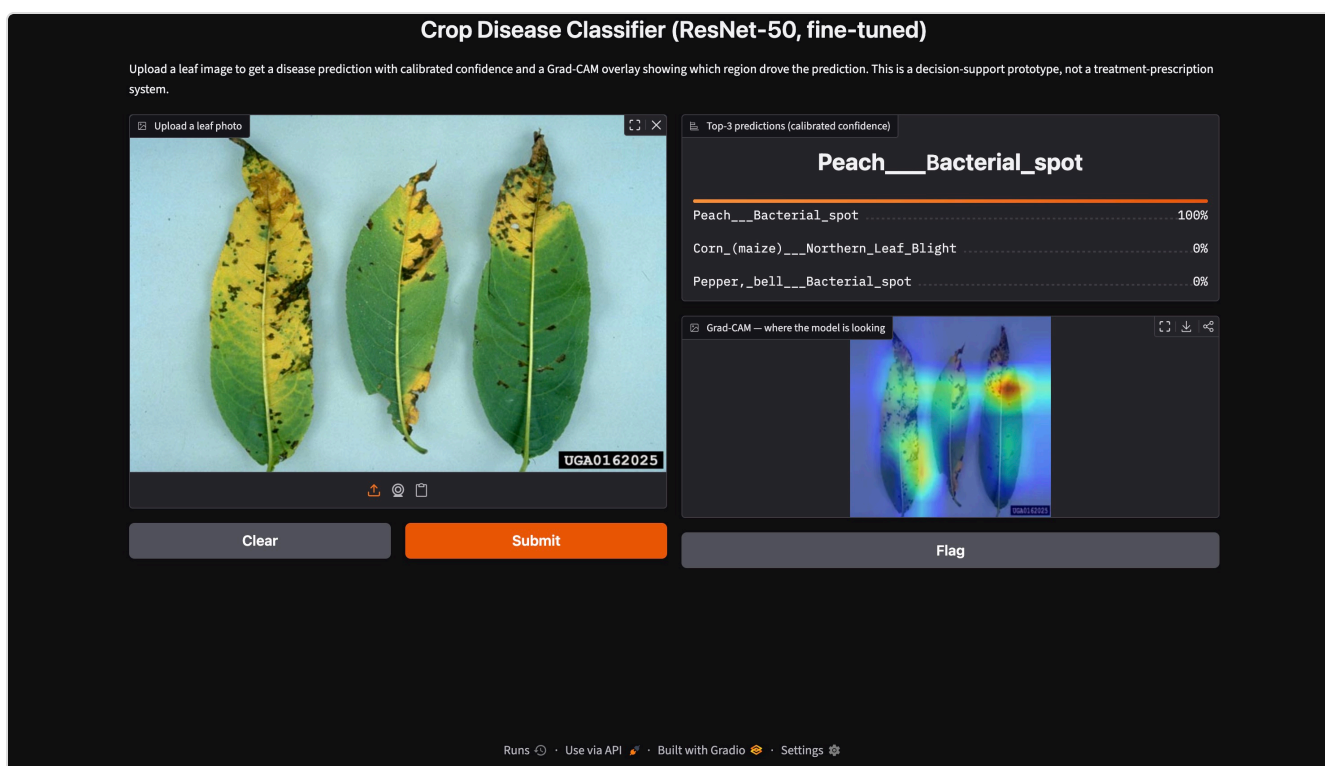


Figure 6 — Real field photo, verified true label: **Peach bacterial spot** (IPM Images #UGA0162025). Correctly classified at 100% confidence, with the heatmap concentrated on the dark bacterial lesions across all three leaves in frame.

Interpretation. This 50% field accuracy (vs. 99.4% on the PlantVillage test set) is consistent with the background-bias risk identified in the proposal: PlantVillage's uniform studio backgrounds (plain black, white, or gray backdrops) may have allowed the model to partially key on background texture rather than relying solely on disease-lesion features — visible directly in Figure 2, where the misclassified apple scab photo itself uses a studio-style backdrop closely resembling PlantVillage's own style, yet was still misclassified. Field photographs with natural backgrounds, soil, and foliage fall outside this learned distribution more broadly. Critically, the confidence-calibration mechanism does not reliably degrade under this distribution shift in every case; three of the six predictions above remained confidently wrong (81–99%) rather than appropriately uncertain, though notably the correct citrus greening prediction (Figure 5, 71%) does show the model can express lower confidence appropriately on harder cases.

Deployment implication: consistent with the project's stated position, this reinforces that the system must be deployed strictly as **decision support**, not autonomous diagnosis, and that confidence scores alone cannot yet be fully trusted as a reliable escalation signal for out-of-distribution field images.

Recommended future work

- **Background-replacement augmentation** — the dataset's pre-segmented leaf masks could be composited onto varied real-world backgrounds during training, directly targeting the suspected background-keying behavior.
- **Fine-tuning on real field imagery** — supplementing training with a field-condition dataset such as PlantDoc (2,598 real-world images across overlapping crop/disease categories).
- **Larger-scale field evaluation** — this test used a modest, illustrative sample ($n=6$); a statistically robust assessment would require dozens of verified field images per class.
- **Out-of-distribution detection** — a mechanism to flag inputs that fall outside the training distribution, rather than relying solely on softmax confidence.

7. Technical Deployment Details

7.1 Deployment Overview & Architecture

- **Model format:** PyTorch serialized state dictionary (.pth)
- **Model architecture:** ResNet-50 (ImageNet-pretrained backbone, fine-tuned), 23,585,894 total parameters
- **Artifact file:** new_saved_models/resnet50_unfrozen_best_state_100p.pth (90.3 MB, verified on disk)
- **Inference engine:** PyTorch CPU runtime (no GPU required for inference)
- **Interface:** Gradio local web demo (app.py) — a single-process local application accessed via browser, not a REST API

System flow

```
[User] -> uploads leaf photo via Gradio browser UI
      |
[PIL.Image] -> convert to RGB -> torchvision.transforms (resize 224x224, ToTensor,
ImageNet normalize)
      |
[ResNet-50 model] -> load_state_dict(.pth) -> model.eval() -> forward pass
(torch.no_grad())
      |
[Post-processing] -> softmax(logits / T) with calibrated temperature T=0.615 ->
top-3 classes
      |
[Grad-CAM] -> hook on layer4[-1] -> backward pass on predicted class -> heatmap
overlay
      |
[Gradio UI] -> displays top-3 predictions + confidence + Grad-CAM image
```

7.2 Configuration & Python Dependencies

Python version: 3.12.12 (verified) | CPU-only PyTorch is sufficient — verified to load and run inference correctly with no GPU present.

```
torch==2.13.0
torchvision==0.28.0
pillow==12.3.0
```

```
matplotlib          # required for Grad-CAM colormap rendering
gradio
numpy
```

Genuine finding during verification: the first deployment attempt failed with `ModuleNotFoundError: No module named 'matplotlib'` — `matplotlib` is required for the Grad-CAM heatmap colormap and must be included in the dependency list above; it is easy to omit since the main prediction path does not need it.

7.3 Operational Verification

actually executed, not simulated

Test	Input	Result	Details
1 — Valid image	Apple___healthy sample	✅ SUCCESS	Predicted: Apple___healthy, confidence 1.0000, inference time 0.0313s, model load time 0.387s
2 — Invalid input	Non-image .txt file	⚠️ ERROR (expected, caught)	<code>PIL.UnidentifiedImageError: cannot identify image file</code>
3 — Corrupted image	Truncated/garbage .jpg bytes	⚠️ ERROR (expected, caught)	<code>OSError: Truncated File Read</code>

Implication: the demo must wrap image loading in a try/except and show the user a clear "please upload a valid image" message for both invalid and corrupted uploads, since PIL raises these errors rather than failing silently or crashing the whole app.

7.4 Deployment Risk & Mitigations

- **Memory:** `torch.no_grad()` is used during the main inference pass to avoid building a gradient graph; Grad-CAM separately requires gradients only for its own scoped backward hook.
- **Error handling:** confirmed above — invalid/corrupted image uploads raise catchable exceptions rather than crashing silently.
- **Reproducibility risk:** the fitted temperature ($T=0.615$) and checkpoint are tied to this specific training run — if the model is retrained, both the checkpoint path and `TEMPERATURE` constant must be updated together, or confidence scores will be miscalibrated.

8. Summary Metrics (Technical Appendix)

Metric	Value
Model	ResNet-50 (ImageNet-pretrained, fine-tuned)
Held-out test accuracy	99.4%
Held-out test macro-F1	0.991
Calibration (ECE)	0.049 → 0.004 after temperature scaling ($T = 0.615$)
Field-photo accuracy (verified sample, n=6)	50% (3/6 correct), three high-confidence errors
Minimum training data for target macro-F1 (≥ 0.88)	Achieved with as little as 15% of available training data
Total model parameters	23,585,894
Checkpoint file size	90.3 MB